

Dipayan Chakraborty

physics.dipayan@gmail.com

+402 735209802, +420 266053503 (Land), +91 8972380308 ORCID Id: 0000-0002-4154-3314

Professional Experiences/Affiliations:

- **Associate Scientist** *February, 2026 – Present*
J. Heyrovský Institute of Physical Chemistry of the CAS, Prague, Czech Republic
Supervisor: Dr. Jaroslav Kočíšek
- **Research Associate** *July, 2025 – October, 2025*
Indian Institute of Science Education and Research Kolkata, India
Supervisor: Prof. Dhananjay Nandi
- **Postdoctoral Research Associate** *June, 2019 – April 2025*
Radiation Laboratory, University of Notre Dame, Indiana, United States
Supervisor: Prof. Sylwia Ptasinska
- **Affiliated Research Scholar** *September, 2021 – May 2024*
Lawrence Berkeley National Laboratory, Berkeley, California, United States
Supervisor: Dr. Daniel S. Slaughter
- **Visiting Scientist** *September, 2021 – October, 2021*
Lawrence Berkeley National Laboratory, Berkeley, California, United States
Supervisor: Dr. Daniel S. Slaughter

Education

- **PhD:** Experimental Atomic and Molecular Physics *2019*
Indian Institute of Science Education and Research Kolkata, India
PhD Supervisor: Prof. Dhananjay Nandi
- **MS:** Physics *2014*
National Institute of Technology, Durgapur, India
- **BS:** Physics *2011*
The University of Burdwan, India

List of Publications

Published as a lead author and/or corresponding author (*)

1. **Dipolar dissociation dynamics in electron collisions with carbon monoxide**
Dipayan Chakraborty, Pamir Nag and Dhananjay Nandi
Phys. Chem. Chem. Phys., 18, 32973, (2016)
2. **A new time of flight mass spectrometer for absolute dissociative electron attachment cross-section measurements in gas phase**
Dipayan Chakraborty, Pamir Nag and Dhananjay Nandi
Rev. Sci. Instrum., 89, 025115, (2018)
3. **Dissociation dynamics in low energy electron attachment to ammonia using velocity slice imaging**
Dipayan Chakraborty, Aranya Giri and Dhananjay Nandi
Phys. Chem. Chem. Phys., 21, 21908, (2019)

4. **Absolute dissociative electron attachment cross-section measurements of difluoromethane**
Dipayan Chakraborty and Dhananjay Nandi
[Phys. Rev. A](#), 102, 052801, (2020)
5. **Dissociative electron attachment to amide bond containing molecules: N-ethylformamid and N-ethylacetamide**
Dipayan Chakraborty Lauren Eckermann, Ian Carmichael and Sylwia Ptasinska
[J Chem. Phys.](#), 153, 224306, (2020)
6. **Topical Review: Dissociation dynamics of ion-pair states accessed by low-energy electron collisions**
Dipayan Chakraborty* and Anirban Paul
[J. Phys. B: At. Mol. Opt. Phys.](#), 56, 142001, (2023)
7. **Dynamics of resonant low-energy electron attachment to ethanol-producing hydroxide anion.**
Dipayan Chakraborty* Daniel S. Slaughter, and Sylwia Ptasinska,
[Phys. Rev. A](#), 108, 052806, (2023)
8. **Dissociative electron attachment to gas-phase acetic acid using a velocity map imaging technique.**
Dipayan Chakraborty Giorgi Kharchilava, Ian Carmichael, and Sylwia Ptasinska
[J. Phys. B: At. Mol. Opt. Phys.](#), 56, 245202, (2023)
9. **Structural rearrangements and fragmentation pathways induced by a low-energy electron attachment to ethyl acetate.**
Anirban Paul, Ian Carmichael, Dhananjay Nandi, Sylwia Ptasinska, and Dipayan Chakraborty*
[Phys. Rev. A](#), 109, 042818, (2024)
10. **Dissociation of gas-phase anisole induced by low-energy electron interactions: understanding patterns of aromatic bond cleavage**
Jacob Finley, Giorgi Kharchilava, Ian Carmichael, Dipayan Chakraborty*, and Sylwia Ptasinska
[Phys. Scr.](#), 100, 045402, (2025)
11. **Fragmentation dynamics through geometrical distortion in low-energy electron attachment to carbon disulfide**
Anirban Paul, Daniel S. Slaughter, Sylwia Ptasinska, Dhananjay Nandi, Ian Carmichael, Jimena D. Gorfinkiel, and Dipayan Chakraborty*
[Phys. Rev. A](#), 111, 042804, (2025)

Other co-authorship papers

1. **Low-Energy Electron Generation for Biomolecular Damage Inquiry: Instrumentation and Methods**
Elahe Alizadeh, [Dipayan Chakraborty](#), and Sylwia Ptasinska
[Biophysica](#), 2, 475, (2022)
2. **Low-energy electron driven reactions in 2-bromo-5-nitrothiazole**
Jiakuan Chen, [Dipayan Chakraborty](#), Milan Oncák, Sylwia Ptasinska, and Stephan Denifl,
[J. Chem. Physics](#), 162, 044304, (2025)
3. **Study of low-energy electron-induced dissociation of 1-propanol**
Soumya Ghosh, [Dipayan Chakraborty](#), Anirban Paul, and Dhananjay Nandi
[J. Chem. Physics](#), 164, 054302, (2026)

Conference publications

1. **Dipolar dissociation dynamics in electron collisions with carbon monoxide**
[Dipayan Chakraborty](#), Pamir Nag and Dhananjay Nandi
[Journal of Physics: Conf. Series](#), 875, 062008, (2017)

Reviewer of International Journals

Web of Science ResearcherID: JQW-4590-2023

- The Journal of Chemical Physics. (AIP)
- Journal of Physics B: Atomic, Molecular, and Optical Physics (IOP)
- Measurement Science and Technology (IOP)
- Journal of Physics D: Applied Physics (IOP)
- Physica Scripta (IOP)

Academic Awards and Fellowships:

- Department of Energy Postdoctoral Research Fellowship, Radiation Laboratory, University of Notre Dame. June 2019
- Secured Joint Entrance Screening Test (JEST- 2014) conducted by the Department of Energy, the government of India.
- Indian Institute of Science Education and Research (IISER) Kolkata Ph.D. research fellowship (2014 – 2019).
- PhD thesis was shortlisted for the best thesis award (Top 4) by the Indian Society of Atomic and Molecular Physics (ISAMP) for the year 2018-2020.
- Received International Student Travel Grant from IISER Kolkata in February 2018.

Supervision Activities:

Consultant of MS theses of Lauren Eckermann (completed), Giorgi Kharchilava, and Jacob Finley (both ongoing) in Radiation Laboratory, University of Notre Dame.

Organizing Activities:

Local organizing committee member of the following conferences

- [POSMOL 2021](#) (University of Notre Dame, United States, 29 – 30 July 2021)
- [POSMOL 2023](#) (University of Notre Dame, United States, 3 – 6 August 2023)

Presentation and talks delivered at important conferences:

1. **Development of a velocity map imaging spectrometer and its application in understanding the dynamics of low energy electron-molecule collisions.** ([Contributed talk](#))
27th International Conference on the Application of Accelerators in Research and Industry (CAARI) and the 55th Symposium of Northeastern Accelerator Personnel (SNEAP) July 21-26, 2024, Fort Worth, Texas.
2. **Dynamics of resonant low energy electron attachment to dimethyl formamide.** ([Contributed talk](#))
Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP), June 3–7, 2024; Fort Worth, Texas., abstract id. C11.00003

3. **Dynamics of resonant low-energy electron attachment to ethanol-producing hydroxide anions. (Invited talk)**
 POSMOL 2023, University of Notre Dame, Notre Dame in Indiana, USA, on August 3-6, 2023
4. **Dissociation dynamics of low energy electron attachment to carbon disulfide studied using velocity slice imaging technique. (Contributed talk)**
 APS Gaseous Electronics Conference 2023, University of Michigan, Ann Arbor, Michigan, USA, on October 9-13, 2023, abstract id. FW4.00003
5. **Dissociative electron attachment studies of molecules containing amide bonds. (Contributed talk)**
 APS Gaseous Electronics Conference 2020, Virtual, October 5, 2020, abstract id.FT1.003
6. **Signature of A1 symmetry in the higher resonance state in dissociative electron attachment to Ammonia. (Poster presentation)**
 Asian International Seminar on Atomic and Molecular Physics (AISAMP 13), TIFR Mumbai and IIT Mumbai, Mumbai India, 3-8 December 2018
7. **A new time of flight mass spectrometer for absolute dissociative electron attachment cross-section measurements. (Selected as hot topic presentation)**
 3rd International Workshop on Dissociative Electron Attachment Prague, Czech Republic, 10-13th April 2018.
8. **Absolute Dissociative Electron Attachment Cross-section Measurement Studies for Difluoromethane. (Poster presentation)**
 Conference on Atomic and Molecular Physics (ISAMP-TC7), IISER-Tirupati and IIT-Tirupati, Tirupati, India, 6-8 January 2018.
9. **Electron Impact Ionization and Dissociation of Isolated Molecules and Clusters. (Poster presentation)**
 21st National Conference on Atomic and Molecular Physics (NCAPM-XXI), Physical Research Laboratory, Ahmedabad, India, 3-6 January 2017.
10. **Polar Dissociation Dynamics in Electron Collisions With Carbon Monoxide. (Poster presentation)**
 International Conference on "Charged Particle Collisions and Electronic Processes in Atoms, Molecules, and Materials", q-PaCE 2016, Indian School of Mines, Dhanbad, India, 9-11 January 2016.

Key Technical skills

- **Designing and commissioning of new experiments:** Designed and constructed velocity map imaging spectrometers and absolute electron attachment cross-section measurement setup completely from scratch to study low-energy collision processes in the gas phase.
- **Data acquisition and interfacing:** Extensively used NI LabVIEW to develop the data acquisition and instrument control system for the velocity map imaging set up at the Radiation Laboratory, University of Notre Dame, during my Postdoc.
- **Programming and data analysis:** Experienced Matlab programmer, wrote all the data analysis codes for the VMI experiment in the Radiation Laboratory, University of Notre Dame.
- **Computer-aided Designing (CAD) software:** Experienced Autodesk inventor and SolidWorks CAD software user. Used Autodesk Inventor software to design several experimental setups.
- Experienced SIMION, ion trajectory simulating software user.

- I have extensive experience in designing, working with, and maintaining high- and ultra-vacuum systems for more than nine years.
- Operating systems: Comfortable with Windows, and macOS operating systems.

Research Experience

- I obtained a PhD degree from IISER Kolkata, India, under the supervision of Prof. Dhananjay Nandi. The primary focus of the thesis work was to develop a new spectrometer for measuring the absolute dissociative electron attachment cross-section of biologically important molecules. The title of my PhD thesis was “Low-energy Electron-Molecule Collision Studies in Gas Phase.” I developed the setup, which included the installation of the microchannel plate (MCP), vacuum components, and vacuum pumps. Additionally, I installed all data acquisition electronics and used SIMION simulations to optimize the voltage of the spectrometer electrodes and the instrument control system. Subsequently, I successfully tested and utilized the setup to study various molecules.
- During my PhD, I also had the opportunity to conduct experiments using a velocity map imaging spectrometer. I investigated two fundamental electron-molecule collision phenomena: dissociative electron attachment and ion-pair dissociation.
- During my postdoc tenure at the University of Notre Dame, I employed a Quadrupole Mass Spectrometer (QMS) in conjunction with an electron monochromator to investigate resonant low-energy electron collisions with amide bond-containing molecules. Such studies have significant applications in the field of medical science.
- Since the beginning of 2021, I have been employed to build a Velocity Map Imaging Spectrometer at the Radiation Laboratory, University of Notre Dame. I developed the entire setup completely from scratch, involving design, construction, simulation, development of the data acquisition system, writing data analysis codes, spectrometer voltage calibration, and ultimately conducting experiments to generate publishable results.

Teaching Experiences:

- Teaching Assistant for Physics I course (course for 1st-year undergraduate level students) in autumn 2015 at IISER Kolkata.